ADDENDUM TO "EVEN UNIMODULAR IDEAL LATTICES OF RANK 88 OVER $\mathbb{Q}(\zeta_{115})$": TWO FURTHER EXTREMAL LATTICES, AN ADVERSARIAL AUDIT, AND THE TWINS OF THE σ_{24}-LAYER

IVAN GAUVIN

ABSTRACT. This addendum to [Gau26] has three parts. (1) The ideal lattices X_{4969} and X_{4049}, which [Gau26] listed as partial survivors of a filter, are extremal: their minimum 8 is established by the computer-assisted method of [Gau26], each of their four sides having been swept twice, with byte-identical receipts. Without assuming GRH, their ideal classes are conjugate neither to each other nor to those of X_{139}, X_{599} and of Sakamoto's two lattices; by the isometry theorem of [Gau26], six orbits of extremal lattices of rank 88 are now known among the unimodular ideal lattices over $\mathbb{Q}(\zeta_{115})$. (2) The certified enumeration was attacked from seven sides: predicted hits, mutations, an independent recount, replications, a generic control group, a change of representative, and a review by a second assistant. (3) The 32 orbits fixed by σ_{24} are carried by the powers of a single prime ideal, P_{139}^k, and the orbit of an explicit ideal of this layer can be read without GRH; Sakamoto's lattice L_S is isometric to the lattice of P_{139}^{331}. These orbits form sixteen *twin pairs*, whose members differ by the class of Sakamoto's ideal. Twin certificates without GRH show that X_{4049} and X_{599} are twins, like L_0 and L_S, whereas the twin of X_{4969} contains a vector of norm 6: extremality is not an invariant of twin pairs. Overnight filters on 21 of the 22 undecided orbits of the layer rejected 4 of them and left 17 partial survivors, far more than a naive heuristic predicts.

This addendum uses the notation and the five statuses of [Gau26]: [proved], [under GRH], [computer-assisted], [observed], [open], and [heuristic] for a heuristic. A statement [computer-assisted] rests on the execution model of [Gau26, §5.4] (M1, M2, H1–H3). Numbered references such as [Gau26, Prop. 2.5] are to the English version of the preprint.

1. SUMMARY

Theorem A (six extremal orbits)**.** *The lattices X_{139}, X_{599}, X_{4969} and X_{4049} are even unimodular of rank 88 and minimum 8. They are pairwise non-isometric, and isometric neither to L_0 nor to L_S. Together with Sakamoto's lattices, they form six pairwise distinct orbits of extremal lattices in the set S of* [Gau26].

[minimum of X_{4969}, X_{4049}: computer-assisted (M1, M2, H1–H3), as for X_{139}, X_{599} in [Gau26, Thm. 6.11]; non-isometry: proved, without GRH, as in [Gau26, Thm. 6.11], by non-conjugacy of the classes and the isometry theorem; minimum of L_0, L_S: by [Sak26]]

Theorem B (extremality is not a twin invariant)**.** *In the σ_{24}-layer of S (§4), the twin pairs (L_0, L_S) and (X_{4049}, X_{599}) consist of two extremal lattices, whereas X_{4969} is extremal and its twin, the lattice of the ideal $P_{691}\sigma_{24}(P_{691})$, contains a vector of norm 6.*

[twin relations and the vector of norm 6: proved, without GRH; minima of X_{4969}, X_{4049}, X_{599}: computer-assisted; of L_0, L_S: by [Sak26]]

Theorem C (the dial without GRH)**.** *The 32 orbits of S fixed by σ_{24} are those of the lattices $X_{P_{139}^k}$, $k \in \mathbb{Z}/993$, and they form 16 twin pairs. The orbit of an explicit ideal of this layer can be determined without GRH; in particular, the 30 non-central orbits are represented by ideals above the primes listed in Table 3, of norm at most $53\,269^2$.*

Date: October 17, 2026.

2020 *Mathematics Subject Classification.* Primary 11H06; Secondary 11H56, 11R18, 11R29.

Key words and phrases. extremal lattices, even unimodular lattices, ideal lattices, cyclotomic fields, class group, certified enumeration.

[proved, without GRH, modulo [Gau26, Prop. 2.2, 2.5]]

Theorem A is proved in §2, Theorems B and C in §4 (Propositions 4.1, 4.7 and Corollaries 4.2, 4.8). Question (Q2) of [Gau26] is updated: at least six of the 508 463 orbits of S consist of extremal lattices. Question (Q3) is answered for two of the four partial survivors.

New statement	Status
X_{4969}, X_{4049}: even unimodular; vector of norm 8; explicit subgroup of order 460 of the automorphism group (Prop. 2.1)	proved (exact checks)
X_{4969}, X_{4049}: no non-zero vector of norm ≤ 6 (Prop. 2.2)	computer-assisted (H1–H3); each side executed twice, on two machines, identical receipts
Classes of P_{4969}, P_{4049}: order 993, stabilizer $\langle\sigma_{24}\rangle$; non-conjugacy to $[P_{139}]$, $[P_{599}]$ and to each other (Prop. 2.3)	proved, without GRH
Audit of the method (§3)	each item with its own status
The σ_{24}-layer is generated by $[P_{139}]$; $L_S \cong X_{P_{139}^{331}}$ (Prop. 4.1)	proved, without GRH (modulo [Sak26] for L_S)
Twin pairing (Cor. 4.2)	proved
Twin and same-orbit certificates (Prop. 4.5)	proved, without GRH
$X_{P_{691}\sigma_{24}(P_{691})}$ contains a vector of norm 6 (Prop. 4.6)	proved, without GRH
Labels and hours of the 30 orbits of the layer (Prop. 4.7, Cor. 4.8)	proved, without GRH
Filters of the night on 21 orbits of the layer: 4 rejections, 17 partial survivors (§4.5)	rejections proved, without GRH; survivors observed
Enrichment of extremality in symmetric orbits (§4.6)	heuristic

2. Two further extremal lattices

2.1. **The ideals.** The primes 4969 and 4049 are congruent to 24 modulo 115. The prime ideals

$$P_{4969} = (4969,\ \zeta^2 + 3763\zeta + 2809), \qquad P_{4049} = (4049,\ \zeta^2 + 105\zeta + 2084)$$

have residue degree 2, conjugates $\bar{P}_{4969} = (4969, \zeta^2 + 3189\zeta + 1493)$ and $\bar{P}_{4049} = (4049, \zeta^2 + 1669\zeta + 1327)$, and absolute norms $q = 4969^2 = 24\,690\,961$ and $q = 4049^2 = 16\,394\,401$. Their classes are fixed by σ_{24}, the Frobenius substitution of p. We set $X_p = (P_p, \beta_0/\nu)$ as in [Gau26, §6]. In [Gau26, §6.3], X_{4969} and X_{4049} appear as partial survivors of a filter, after 3.8% and 5.4% of the tasks of one side.

Proposition 2.1. *For $p \in \{4969, 4049\}$:*

1. *X_p is a positive definite even unimodular lattice of rank 88, and the input files of the certifying program are exactly linked to it, for both sides;*
2. *X_p contains a vector of norm 8;*
3. *the multiplication g_1 by $-\zeta$ and a semilinear map $g_2 \colon u \mapsto \lambda\sigma_{24}(u)$, with λ a unit of $\mathbb{Z}[\zeta]$, generate a subgroup of order 460 of $\mathrm{Aut}(X_p)$, with $g_2^2 = -1$ and $g_2^{-1} g_1 g_2 = g_1^{139}$.*

[proved]

Justification. (1) The verifier of the geometric link of the repository of [Gau26], the generic form of the one used for X_{139} and X_{599} [Gau26, Prop. 6.3, 6.4], works in integers and rationals, without PARI/GP; it reports 37 checks and no failure for each p. It checks that the basis generates exactly P_p (index p^2, stability under ζ, $P_p = (p, g(\zeta))$ with g irreducible of degree 2 modulo p, $P_p \neq \bar{P}_p$); that the Gram matrix, recomputed from α, is integral, even, of

determinant 1 and positive definite; that ν generates $\mathrm{N}(P_p)$; the gluing identities of [Gau26, Prop. 4.1]; for each side, the bijection between P_p and the pairs (u, y) and the identities that bind the four input files to the lattice; and that these four files are those whose hashes the first execution printed. The identity $\alpha\nu = \beta_0$, which this verifier took for granted, is checked by a separate verifier on the four lattices X_{139}, X_{599}, X_{4969}, X_{4049} (8 checks, no failure; see §3.7).

(2) A short run of the certifying program on side 1, at the bounds $B_A = 29/10$ and $S = 16$ and stopped at the first hit, gives pairs (u, y) of energies

$$\frac{14\,083}{4969} + \frac{1\,300\,307\,796}{9938^2} = 16 \quad (p = 4969), \qquad \frac{10\,194}{4049} + \frac{884\,139\,640}{8098^2} = 16 \quad (p = 4049).$$

The link verifier of (1) checks that the reconstructed vectors have integer coordinates in the basis of the ideal and norm exactly 8.

(3) The unit λ solves $\alpha\lambda\bar{\lambda} = \sigma_{24}(\alpha)$; it was found by floating-point lattice reduction among explicit cyclotomic units, then checked exactly. The relations between g_1 and g_2 and the order 460 were checked in exact arithmetic in PARI/GP, and again by two Python programs in integers and rationals (the scripts accompanying the data of X_{139} and X_{599}, unchanged), which rebuild the Gram matrices and the generators from P_p, α, λ and the basis. □

2.2. **The certified sweeps.** The program `Rang88CertZ` of [Gau26, §5], with the same compiled classes, was run on both sides of each ideal, with $B_A = 1057/200$ and $S = 12$, on rented machines with 32 virtual cores (Scaleway, Ubuntu 24.04), from October 15 to 16, 2026. Each side was executed in a single launch, without resumption, then a second time on another machine (Table 1). The parameters (D_A, D_B, d) are $(4969, 1, 9938)$ and $(4049, 1, 8098)$.

TABLE 1. Certified sweeps of X_{4969} and X_{4049}. Every execution: a single launch with a fresh checkpoint file, exit code 0, empty error stream, complete coverage, no hit, no hit file. The canonical hash (SHA-256 of the receipts sorted by index) is the same for the two executions of each side, and so is the printed $|\mathcal{U}|$ with its histogram.

side	first execution	replication	tasks	printed $\lvert\mathcal{U}\rvert$	canonical hash
X_{4969}, 1	AMS3, Java 21, 31 240 s	WAW1, Java 11, 31 981 s	97 270	1 544 898 536	`0226b9a5...`
X_{4969}, 2	AMS2, Java 21, 31 312 s	AMS3, Java 17, 29 948 s	140 082	1 544 898 536	`f6a1aecd...`
X_{4049}, 1	MIL1, Java 21, 47 775 s	AMS2, Java 21, 22 286 s	77 819	1 257 946 084	`6334d93e...`
X_{4049}, 2	MIL1, Java 21, 48 258 s	AMS2, Java 21, 25 771 s	87 731	1 257 946 084	`f95f16b8...`

Zones: Scaleway availability zones (Amsterdam 2 and 3, Milan 1, Warsaw 1). Java: OpenJDK 21.0.12.1, 17.0.20.1 and 11.0.32.1. Threads: 32, except 15 and 17 for the first executions of X_{4049}. Durations: printed by the program.

For each of the eight executions, a read-only verifier, separate from the program and written for this addendum, rechecks: one launch with a fresh checkpoint file, complete selection, no resumption, the lines `TOTAL_TOUCHES 0` and `COUVERTURE_COMPLETE`, exit code 0 and empty error stream; the hash of the task catalogue, against the log and the receipt header; the strict format of the receipts, the permutation of their indices, and `TOUCHES=0` on each of them; the agreement of their sums, histogram included, with the printed totals; the hash of the receipt file; the absence of hit files; the canonical hash, computed in two ways; and, for the replications, the byte-for-byte equality of each receipt with the corresponding receipt of the first execution. All checks pass (27 to 29 per execution). The four replications reproduce all 402 902 receipts, without a single difference, under three versions of the Java virtual machine. The first executions also reproduce, byte for byte, the receipts of earlier partial runs on two other computers (3 716 receipts for side 1 of X_{4969}, 8 517 and 4 594 for the two sides of X_{4049}).

The enumeration trees are large: side 1 of X_{4969} alone visits $1.07 \cdot 10^{12}$ nodes in rank 44 for the short projections and $1.25 \cdot 10^{13}$ in the fibers.

Proposition 2.2. *X_{4969} and X_{4049} contain no non-zero vector of norm at most 6.* [computer-assisted]

Proof. For each of the four sides, Corollary 5.2 of [Gau26] applies to the fresh executions of Table 1: exit code 0, the line `COUVERTURE_COMPLETE`, no hit file in the output directory (complete inventory taken after the end of the computation), and launch scripts that delete nothing. Hence $\mathcal{T} = \emptyset$ for each side, under H1–H3, and the criterion [Gau26, Thm. 4.6], with the geometric link of Proposition 2.1, gives the claim. Hypothesis H4 plays no role. □

2.3. **Classes and non-isometry, without GRH.** Let $K = K_{22} = \mathbb{Q}(\theta)$, $\theta = \zeta + \zeta^{22} + \zeta^{24} + \zeta^{68}$, as in [Gau26, §6.4], and $\mathfrak{q}_p = P_p \cap K$; it has degree 1, and $\mathrm{N}_{L/K}(P_p) = \mathfrak{q}_p^2$.

Proposition 2.3. *For $p \in \{4969, 4049\}$:*

(a) *$[P_p]$ has order 993; its support is $\{3, 331\}$, its stabilizer is $\langle \sigma_{24} \rangle$, and its orbit has size 44;*

(b) *$[P_p]$ is conjugate under $\mathrm{Gal}(L/\mathbb{Q})$ neither to $[P_{139}]$, nor to $[P_{599}]$, nor to the class of the other ideal.*

[proved, without GRH, modulo [Gau26, Prop. 2.2, 2.5]]

Proof. A certificate, verified by a Python program in exact arithmetic without PARI/GP, establishes the following, in the notation of [Gau26, §6.4]: an explicit $\Gamma_p \in K$ with $(\Gamma_p) = \mathfrak{q}_p^{331}$, $\mathfrak{q}_{4969} = (4969, \theta - 4770)$, $\mathfrak{q}_{4049} = (4049, \theta - 512)$ (norm p^{331}, valuations at the 22 primes above p); the rank 10 of the discrete logarithms modulo 331 of the units $\xi_2, \dots, \xi_{11}$ at the 88 primes of degree 1 above $\ell \in \{380\,651;\ 1\,065\,821;\ 1\,522\,601;\ 2\,055\,511\}$, all $\equiv 1 \pmod{331 \cdot 115}$; the rank 11 once the column of Γ_p is added; and, for each of the 22 elements τ of $\mathrm{Gal}(K/\mathbb{Q})$, the rank 11 for $\Gamma_p/\tau(\Gamma_{p'})$, with $p' = 139$, and also $p' = 4969$ when $p = 4049$. The verifier reports 39 checks for P_{4969} and 49 for P_{4049}, no failure; they include an autonomous check that the basis of the input files generates P_p (index p^2).

(a) The rank 11 and Steps 1–3 of [Gau26, §6.4] show that P_p^m is not principal for m prime to 331: the 331-component is non-zero. The prime p is $\equiv 1 \pmod{23}$, hence splits in $F = \mathbb{Q}(\sqrt{-23})$, and $4p$ is not of the form $a^2 + 23b^2$ ($4 \cdot 4969 = 19\,876$, $4 \cdot 4049 = 16\,196$): the prime $\mathfrak{p}$ of F above p is not principal, nor is $\mathrm{N}_{L/F}(P_p) = \mathfrak{p}^2$, and the 3-component is non-zero, by the argument of [Gau26, Rem. 6.8(a)]. The 45 013-component vanishes because σ_{24} fixes P_p and does not lie in the kernel of $\chi_{45\,013}$. As in [Gau26, Rem. 6.8(a)], the order is 993, the stabilizer is $\ker \chi_3 \cap \ker \chi_{331} = \{1, 24\}$ and the orbit has size 44.

(b) If $[P_p] = \sigma[P_{p'}]$ for some $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$, then, taking norms to K, $\mathfrak{q}_p^2\,\tau(\mathfrak{q}_{p'})^{-2}$ would be principal, with $\tau = \sigma|_K$, and $\Gamma_p^2/\tau(\Gamma_{p'})^2$ would lie in $E \cdot K^{\times 331}$; Step 1 and the rank 11 forbid this, exactly as in Step 2 of [Gau26, §6.4]. Finally $[P_{599}]$ has zero 3-component [Gau26, Prop. 6.7], whereas $[P_p]$ has not. □

Proof of Theorem A. Evenness, unimodularity and the vector of norm 8: Proposition 2.1 and [Gau26, Prop. 6.3, 6.4]. Minimum ≥ 8: Proposition 2.2 and [Gau26, Thm. 6.11]; hence minimum 8. Non-isometry: the classes $[P_{139}]$, $[P_{599}]$, $[P_{4969}]$, $[P_{4049}]$ are pairwise non-conjugate (Proposition 2.3 and [Gau26, Rem. 6.10]), and none of their cubes is trivial, whereas those of the classes of L_0 and L_S are [Gau26, Prop. 2.7]. The isometry theorem [Gau26, Thm. 3.3] applies to lattices of minimum at least 8; all six have minimum 8, so it turns non-conjugacy into non-isometry. As in [Gau26, Prop. 6.9(a)], the non-isometry to L_0 and L_S does not even need the computer-assisted minimum: a lattice of minimum at most 6 is not isometric to a lattice of minimum 8. □

Remark 2.4. By [Gau26, Rem. 2.6] and Proposition 2.3, the subgroup of Proposition 2.1(3) is the full normalizer of the group of multiplications by roots of unity: $|\operatorname{Aut}(X_{4969})| \geq 460$ and $|\operatorname{Aut}(X_{4049})| \geq 460$. Gram matrices in LLL-reduced bases, the two generators and the vectors of norm 8 have been prepared in the format of the Catalogue of Lattices [NS], as for X_{139} and X_{599}.

3. Attacks on the method

A computer-assisted statement is worth the attacks it has survived. On October 16, 2026, the certifying pipeline was attacked deliberately, each attack aimed at one way in which a certified enumeration can fail. The second assistant, Astra, designed part of the plan, reviewed the records, and was asked to look for flaws. Table 2 gives the outcome; each line is detailed below.

Table 2. Seven attacks on the certified enumeration (October 16–17, 2026).

failure mode	attack	outcome
blindness to existing vectors	predict the hits from known witnesses of norm 6 and their 115 rotations, by a PARI/GP script that reads no output of the program; replay only the tasks concerned	334 of 334 predicted hits found, on 7 rejected lattices and 14 sides, plus 22 exact hits of other orbits; one hit $3 \cdot 10^{-5}$ below the threshold
rounding and pruning	8 single mutations of the source (leaf test, budgets, outward roundings, fixed point, zero vector, signs)	3 killed by the tests, 1 without effect; 4 survive every finite test: only the non-omission proof covers them
counting	recount $\lvert\mathcal{U}\rvert$ of X_{139}, side 1, with `qfminim` of PARI/GP	identical at 6 thresholds, including 42 957 972 at 1057/200
machine, hardware	replicate each side of X_{4969}, X_{4049} on another machine (another Java version for X_{4969})	402 902 receipts, 0 differences
dependence on coordinates	sweep a different ideal of the orbit of X_{139}, proved conjugate without GRH	$X_{P_{461}\sigma_{24}(P_{461})} \cong X_{139}$: no hit on either side
generic behaviour	16 degree-1 ideals, filter with stop at the first hit	14 rejected in less than 5 minutes each (3 already rejected by PARI/GP), 2 partial survivors
blind spots of the author	adversarial review by a second assistant	no serious flaw of the preprint; two gaps in the verifiers and a defect in a new verifier, all corrected

3.1. **Sensitivity: predicted hits.** For seven rejected lattices of [Gau26, Table 2] (P_{691}, P_{1151}, P_{1381} of degree 1; P_{829}, P_{1289}, P_{1471}, P_{1979} of degree 2), a PARI/GP script computed, from the published witness z of norm 6 and its rotations $\zeta^k z$, the exact pairs (u, y) of both sides that the criterion must catch, and checked on them the identities of [Gau26, §4] (zero mean, the 23 relations with five terms, Parseval, $\max|f| \geq N\sqrt{5/352}$). It reads no output of the Java program. Each predicted u lies in exactly one task of the catalogue, which is thus checked to be a partition on these vectors. The tasks containing the predicted u were then replayed without stopping at the first hit: the 334 predicted hits were all found, with the same u, y and energies, together with 22 further hits, all of exact norm 6, from other orbits. On side 1 of P_{1151}, a predicted hit of short energy $6083/1151 \approx 5.28497$ lies $3 \cdot 10^{-5}$ below the threshold $1057/200$ (and above the Parseval value $6 - \sqrt{45/88} \approx 5.28490$); it is found.

For the four lattices of degree 2, the starting witness had itself been found by a Java program (by `Rang88CertZ` for three of them); the prediction of the other 114 rotations and of side 2, and the exact check of the extra hits, do not depend on it. In a boundary variant, where B_A is set exactly to the energy of the highest predicted hit of a side, the reference program finds all hits, including the one on the boundary. [observed]

3.2. **Mutations.** Eight copies of the source were made, each with a single substitution, and run against three tests: the test harness of the program (910 checks), the validation of [Gau26, §5.2] on P_{461} at threshold 7/2 (697 short projections, 2 hits), and the replay of 12 tasks of X_{4969} against reference receipts; then against the boundary variant above. Three mutants are killed: the leaf test without equality, a fiber budget lowered by one step, and a change of the sign convention. A fourth, which no longer excludes the zero vector at the last level, has no effect: the zero vector is excluded a second time at the leaf. The four others prune as soon as the remaining budget is zero, remove the outward rounding of the remaining budget or of the radius, or shrink an integer range by one unit of 2^{-40} in fixed point. They pass every test, even at the exact boundary. No finite test of this kind can see them; the correctness of the code they modify rests on the non-omission proof alone (lemmas Z3–Z5, outward rounding L3, model M1). For the first of them, the author's argument, a lower bound on the budget, did not hold in general; Astra replaced it by an invariant of the program, namely that the accumulated partial energy never exceeds the current bound (`PL` $\leq$ `borneU`) on any reachable call, so that the value at which the two versions differ is never reached. This experiment does not weaken the proof; it shows where only the proof protects. [observed]

3.3. **An independent recount.** The number $|\mathcal{U}|$ of short projections of side 1 of X_{139} was recounted with `qfminim` of PARI/GP, another program by other authors, on the same integer Gram matrix. For short energies below 3, 7/2, 4, 9/2 and 5, and at most 1057/200, it gives 162, 4 878, 92 734, 1 260 158, 12 651 642 and 42 957 972, exactly the values printed by `Rang88CertZ`; the last count took 11 260 s. `qfminim` is not certifying, but it is an enumerator independent of the Java code. [observed]

3.4. **Replications.** Each side of X_{4969} and X_{4049} was executed twice, on different rented machines, under OpenJDK 21, 17 and 11 (Table 1). The 402 902 receipts of the four replications are byte-identical to those of the first executions. Together with the replications of [Gau26], every side of the four extremal lattices of Theorem A has now been executed twice with identical receipts. These replications support H2 and H3 without proving them; they share the program, and they are not an independent implementation.

3.5. **Change of representative.** The ideal $I = P_{461}\sigma_{24}(P_{461})$, with $P_{461} = (461, \zeta - 5)$, has norm 461^2 and the same orbit as P_{139} (Proposition 4.5(c) below, without GRH): X_I is isometric to X_{139}, but its ideal, basis, Gram matrix, input files and task catalogue are all different. Its two sides were swept completely on October 16 (73 244 and 73 359 tasks, 9 559 s and 8 750 s, 16 threads each, exit code 0, complete coverage, no hit; 27 checks of the verifier per side). The printed $|\mathcal{U}|$ is 143 529 602 on each side, against 42 957 972 for X_{139}: a different tree, the same verdict. This is a second, computer-assisted proof of the minimum of X_{139} (one execution per side). Conversely, the ideals $P_{1381}\sigma_{24}(P_{1381})$, $P_{3221}\sigma_{24}(P_{3221})$ and $P_{5521}\sigma_{24}(P_{5521})$, in the orbits of the rejected P_{1289}, P_{1979} and P_{829} (Proposition 4.5(d)), gave hits within less than nine minutes; the four hit vectors (two for the second ideal) were checked to have exact norm 6.

3.6. **A generic control group.** The sixteen smallest primes $p \equiv 1 \pmod{115}$ after 461 give ideals $(p, \zeta - r)$ of degree 1, a priori in generic orbits of size 88. A filter on side 1 (author's computer, Java 11, 8 threads), stopped at the first hit and capped at 300 s, rejected 14 of them, after 33 to 941 tasks: the three already rejected in [Gau26] by its PARI/GP filter, and eleven new ones. Two, P_{2531} and P_{11731}, are partial survivors; for P_{2531}, the PARI/GP filter

of [Gau26] had also stopped without a decision. Some rejections required about a thousand tasks: a short filter must cover at least a few thousand of them. [observed]

3.7. **Review.** Astra (OpenAI) had the records attacked by separate agents and reread their reports itself, under fixed hashes: the non-omission proof and its link to the code (zero budgets, conversions, underflow, boundaries, overflow, cache, partition of the tasks, hits; 296 local exact comparisons, without running the Java program); the isometry theorem; the geometric links and the class certificates of X_{4969} and X_{4049}; and the records of their four first executions (exit codes, error streams, archives and inventories of the output directories, in which there is no hit file). It made its own exact checks of $\alpha\nu = \beta_0$ on the four lattices and of the fiber identities on the eight sides. Its conclusion: no serious flaw of the preprint was established. It reported two minor points of presentation in the non-omission proof; that the link verifier asserted $\alpha\nu = \beta_0$ without testing it (the identity holds, and is now tested); that one check of the class certificates identified the basis with the prime ideal only through the link verifier (the certificate now contains its own index test); and a real defect in a verifier written during the audit, a Newton identity that counted one term twice, which was corrected and reread. Astra's review does not certify the minimum of X_{4969} and X_{4049}, and does not claim to. It leaves open what [Gau26, §5.4] leaves open: no independent implementation of the enumeration, and H3.

4. The twins of the σ_{24}-layer

4.1. **All σ_{24}-stable lattices are powers of one ideal.** Let $\mathrm{Cl}(L)^{\sigma_{24}}$ be the subgroup of the classes fixed by σ_{24}. By [Gau26, Prop. 2.5], σ_{24} lies in the kernels of χ_3 and χ_{331}, not in that of $\chi_{45\,013}$; hence $\mathrm{Cl}(L)^{\sigma_{24}} = \mathrm{Cl}_3 \times \mathrm{Cl}_{331}$, cyclic of order 993. It contains the classes of the prime ideals of degree 2 above the primes $p \equiv 24 \pmod{115}$, those of the ideals $P\sigma_{24}(P)$, and those of Sakamoto's two lattices. We identify $\mathbb{Z}/993$ with $\mathbb{Z}/3 \times \mathbb{Z}/331$.

Proposition 4.1 (cyclic layer)**.** (a) *The class $[P_{139}]$ generates $\mathrm{Cl}(L)^{\sigma_{24}}$: every σ_{24}-stable element of S is $X_{P_{139}^k}$ for a unique $k \in \mathbb{Z}/993$.*

(b) *For a prime to 115, $\sigma_a[P_{139}]^k = [P_{139}]^{\varepsilon(a)k}$, where $\varepsilon(a) = (\chi_3(a), \chi_{331}(a))$ runs over the whole group $H = \{\pm 1\} \times \mu_{22}$ of $(\mathbb{Z}/3)^\times \times (\mathbb{Z}/331)^\times$. The 32 orbits of S fixed by σ_{24} are the orbits of H on $\mathbb{Z}/993$:*

$$\{0\}, \qquad \{331, 662\}, \qquad 15 \text{ orbits with } 3 \mid k,\ 331 \nmid k, \qquad 15 \text{ orbits with } 3 \nmid k,\ 331 \nmid k,$$

of supports $\emptyset$, $\{3\}$, $\{331\}$, $\{3, 331\}$ and of sizes 1, 2, 22, 44.

(c) *$L_0 \cong X_{P_{139}^0}$ and $L_S \cong X_{P_{139}^{331}} \cong X_{P_{139}^{662}}$.*

[proved, without GRH; (c) modulo [Sak26, Rem. 3.11, Lemma A.9], through [Gau26, Prop. 2.7]]

Proof. (a) $[P_{139}]$ is fixed by σ_{24}, the Frobenius substitution of 139, and has order 993 [Gau26, Rem. 6.8(a)]. The bijection $S \leftrightarrow \mathrm{Cl}(L)$ of [Gau26, Prop. 2.3] gives the second assertion. (b) σ_a acts on Cl_p through $\chi_p(a)$ [Gau26, Prop. 2.5], and the image of χ_{331} is the subgroup μ_{22} of $\mathbb{F}_{331}^\times$. The kernel of $a \mapsto \varepsilon(a)$ is $\ker\chi_3 \cap \ker\chi_{331} = \{1, 24\}$, so its image has order $88/2 = 44 = |H|$. The orbit count follows; it agrees with [Gau26, Table 1]. (c) The principal class is that of L_0. Let $t \neq 0$ be the 3-component of $[P_{139}]$. The class $[P_{139}]^{331}$ has zero 331-component and 3-component $331\,t = t$, since $331 \equiv 1 \pmod 3$: it has support $\{3\}$, and the unique orbit with this support is that of L_S [Gau26, Prop. 2.7(b)]. Conjugate classes carry isometric lattices [Gau26, §2.1]. □

4.2. **Twins.** The projection $k \mapsto k \bmod 331$ is equivariant. It maps an orbit of support $\{3, 331\}$, of size 44, onto an orbit of support $\{331\}$, of size 22, and the fiber of this map above an orbit of support $\{331\}$ contains exactly 44 classes of support $\{3, 331\}$, hence exactly one orbit.

Corollary 4.2 (twin pairs)**.** *The projection induces a bijection between the* 15 *orbits of support* $\{3, 331\}$ *and the* 15 *orbits of support* $\{331\}$*, and pairs the principal orbit with that of* L_S*. Two twin classes differ by a non-zero* 3*-torsion class, that is, by the class of Sakamoto's ideal* $J\mathcal{O}_L$ *or of its conjugate: the twin of* X_I *is* $X_{IJ\mathcal{O}_L}$ *or* $X_{I\bar{J}\mathcal{O}_L}$*. The* 32 *orbits of the layer form* 16 *twin pairs, and* (L_0, L_S) *is the twin pair above the zero class of* Cl_{331}*. Two classes of the layer with the same support are conjugate if and only if their* 331*-components are.* [proved]

Remark 4.3 (twins as twists)**.** By the proof of [Gau26, Prop. 6.7], the extension of ideals $j\colon \mathrm{Cl}(F) \to \mathrm{Cl}(L)$, with $F = \mathbb{Q}(\sqrt{-23})$, is injective, and its image is the 3-torsion of $\mathrm{Cl}(L)$. The twin of X_I is therefore also $X_{I\mathfrak{a}\mathcal{O}_L}$ or $X_{I\bar{\mathfrak{a}}\mathcal{O}_L}$, where $\mathfrak{a} = (2, (1+\sqrt{-23})/2)$ is a prime of F above 2, which is not principal. Since $\mathrm{N}_{L/L^+}(\mathfrak{a}\mathcal{O}_L) = 2\mathcal{O}_{L^+}$, we have $X_{I\mathfrak{a}\mathcal{O}_L} = (I\mathfrak{a}, \alpha/2)$ when $X_I = (I, \alpha)$. Viewed as Hermitian lattices of rank 44 over $\mathcal{O}_F$, twins thus differ by the twist $I \mapsto I \otimes_{\mathcal{O}_F} \mathfrak{a}$ by an ideal of order 3 of $\mathrm{Cl}(F)$. For $I = \mathcal{O}_L$, this is the passage from L_0 to L_S, up to isometry [Gau26, Prop. 2.7]. [proved]

4.3. **Twin certificates without GRH.** Whether two given classes of the layer are twins is a question about 331-components, which [Gau26] could only read under GRH, in $\mathrm{Cl}(K_{22}) \cong \mathbb{Z}/662$ computed by `bnfinit`. A positive answer, however, can be certified without GRH.

Lemma 4.4. *Let* I_a*,* I_b *be ideals of* L *whose classes are fixed by* σ_{24}*, with* $\mathrm{N}_{L/K}(I_a) = \mathfrak{q}_a^2$ *and* $\mathrm{N}_{L/K}(I_b) = \mathfrak{q}_b^2$*. If some* $\delta \in K^\times$ *satisfies* $(\delta) = \mathfrak{q}_a^2\,\tau(\mathfrak{q}_b)^{-2}$ *for some* $\tau \in \mathrm{Gal}(K/\mathbb{Q})$*, then the* 331*-components of* $[I_a]$ *and* $[I_b]$ *are conjugate under* $\mathrm{Gal}(L/\mathbb{Q})$*.*

Proof. The hypothesis gives $\mathrm{N}_{L/K}[I_a] = \tau\,\mathrm{N}_{L/K}[I_b] = \mathrm{N}_{L/K}(\tilde{\tau}[I_b])$ in $\mathrm{Cl}(K)$, for any lift $\tilde{\tau}$ of τ. Let $j\colon \mathrm{Cl}(K) \to \mathrm{Cl}(L)$ be the extension of ideals; $j \circ \mathrm{N}_{L/K} = \sum_{g\in\mathrm{Gal}(L/K)} g$ acts on Cl_{331} by $\sum_g \chi_{331}(g) = 4$, since $\mathrm{Gal}(L/K) = \langle\sigma_{22}\rangle = \ker\chi_{331}$. Hence $4[I_a]_{331} = 4(\tilde{\tau}[I_b])_{331}$, and 4 is invertible modulo 331. □

The element δ is found with `bnfisprincipal`, under GRH; it is then checked without any class group: the maximal order of K is certified (`nfcertify`), $\mathfrak{q}_a$ and $\tau(\mathfrak{q}_b)$ are prime ideals of degree 1 given by two generators, and the equality of ideals $(\delta) = \mathfrak{q}_a^2\tau(\mathfrak{q}_b)^{-2}$ is checked exactly, together with $|\mathrm{N}(\delta)|$. The 3-component of the class of an ideal P of degree 1 or 2 above $p \equiv 1 \pmod{23}$ vanishes if and only if $4p = a^2 + 23b^2$ for some integers a, b; that of $P\sigma_{24}(P)$ is twice that of P.

Proposition 4.5. (a) *X_{4049} and X_{599} are twins.*

(b) *X_{4969} and $X_{P_{691}\sigma_{24}(P_{691})}$ are twins, where $P_{691} = (691, \zeta - 4)$ and $\sigma_{24}(P_{691}) = (691, \zeta - 596)$.*

(c) *$P_{461}\sigma_{24}(P_{461})$ lies in the orbit of P_{139}; hence $X_{P_{461}\sigma_{24}(P_{461})} \cong X_{139}$.*

(d) *$P_{1381}\sigma_{24}(P_{1381})$, $P_{3221}\sigma_{24}(P_{3221})$ and $P_{5521}\sigma_{24}(P_{5521})$ lie in the orbits of P_{1289}, P_{1979} and P_{829}.*

[proved, without GRH, modulo [Gau26, Prop. 2.2, 2.5]]

Proof. For each pair, an element δ as in Lemma 4.4 was found and checked (between 839 and 937 characters as a polynomial in θ), with $\mathfrak{q} = (p, \theta - r)$ and $r = 4770$, 512, 333, 152 for $p = 4969$, 4049, 599, 691; the same $r = 152$ is obtained from P_{691} and from $\sigma_{24}(P_{691})$, as it must, since $\sigma_{24} \in \mathrm{Gal}(L/K)$. The 3-components are read from $4p = a^2 + 23b^2$: none of $4\cdot139$, $4\cdot461$, $4\cdot1289$, $4\cdot1381$, $4\cdot1979$, $4\cdot3221$, $4\cdot4049$, $4\cdot4969$ is of this form, whereas $4\cdot691 = 44^2 + 23\cdot6^2$, $4\cdot829 = 2^2 + 23\cdot12^2$, $4\cdot5521 = 94^2 + 23\cdot24^2$, and $[P_{599}]$ has zero 3-component [Gau26, Prop. 6.7]. Thus the two members of each pair of (c) and (d) have the same support and conjugate 331-components, hence are conjugate (Corollary 4.2); in (a) and (b), the supports are $\{3, 331\}$ and $\{331\}$, and the classes are twins. □

Proposition 4.6. *The lattice X_I, $I = P_{691}\sigma_{24}(P_{691})$, contains a vector of norm* 6*. It is not extremal.* [proved, without GRH]

Proof. A filter of `Rang88CertZ` on the two sides of I (input files prepared as for the other ideals, $D_A = 691$, $d = 1382$), stopped at the first hit, found on the night of October 16 to 17, after 195 and 24 tasks (183 s and 109 s), the pairs of energies

$$\frac{3542}{691} + \frac{13\,129\,000}{1382^2} = 12, \qquad \frac{2406}{691} + \frac{16\,268\,904}{1382^2} = 12.$$

A PARI/GP script independent of the program checks exactly that the basis generates I (its columns lie in both prime ideals, and its determinant is 691^2), that the Gram matrix recomputed from α is integral, even, of determinant 1 and positive definite, and that both vectors have integer coordinates in the basis of I and norm exactly 6 (10 checks, no failure). By [Gau26, Prop. 2.3(b)], X_I is the element of S of class $[I]$. □

Proof of Theorem B. L_0 and L_S are twins by Corollary 4.2, extremal by [Sak26]. X_{4049} and X_{599} are twins by Proposition 4.5(a), extremal by Theorem A. X_{4969} is extremal by Theorem A, and its twin is not, by Propositions 4.5(b) and 4.6. □

4.4. **The dial.** The group $\mathbb{F}_{331}^{\times}/\mu_{22}$ is cyclic of order 15, generated by the class of 2: the number 3 is a primitive root modulo 331 and $\log_3 2 = 121$ is prime to 15. An orbit of support $\{331\}$ or $\{3, 331\}$ is therefore labelled by an *hour* $h \in \mathbb{Z}/15$: the orbit of $[P_{139}]^k$ has hour h when $k \bmod 331 \in 2^h\mu_{22}$. Twins have the same hour, and X_{139} has hour 0. The smallest positive exponents of the hours $0, 1, \ldots, 14$ are $1, 2, 4, 8, 16, 32, 7, 14, 28, 56, 13, 5, 10, 20, 19$. The hour of a given ideal is a discrete logarithm in Cl_{331}. The census of §4.5 first read it in $\mathrm{Cl}(K_{22})$, computed by `bnfinit` under GRH. The following proposition reads it without GRH, with the tools of the class certificates of [Gau26, §6.4]: a generator of $\mathfrak{q}^{331}$, the units $\xi_2, \ldots, \xi_{11}$, and discrete logarithms modulo 331 at auxiliary primes.

Proposition 4.7 (the dial without GRH)**.** *Let $\lambda(x) \in \mathbb{F}_{331}^{88}$ be the vector of the discrete logarithms modulo 331 of an element x of K at the 88 primes of degree 1 of K above $\ell \in \{380\,651;\ 1\,065\,821;\ 1\,522\,601;\ 2\,055\,511\}$ (the auxiliary primes), and U the span of $\lambda(\xi_2), \ldots, \lambda(\xi_{11})$. Let I be an ideal of L with $\mathrm{N}_{L/K}(I) = \mathfrak{q}^e$, where $\mathfrak{q}$ is a prime of K of degree 1, prime to the auxiliary primes, and $e \in \{1, 2\}$; let $\Gamma_{\mathfrak{q}} \in K$ with $(\Gamma_{\mathfrak{q}}) = \mathfrak{q}^{331}$. Then the 331-component of $[I]$ is $c\,[P_{139}]_{331}$, where $c \in \mathbb{F}_{331}$ is the unique solution of*

$$e\,\lambda(\Gamma_{\mathfrak{q}}) \equiv 2c\,\lambda(\Gamma_{139}) \pmod{U}.$$

In particular, if the class of I lies in the σ_{24}-layer and $c \neq 0$, the hour of I is the class of c in $\mathbb{F}_{331}^{\times}/\mu_{22}$. [proved, without GRH, modulo [Gau26, Prop. 2.2, 2.5]]

Proof. The map λ is defined on the elements of K that are units at the auxiliary primes; it is additive and vanishes on ± 1 and on 331st powers. Let E be the unit group of K. By Step 1 of [Gau26, §6.4] (rank 10), $\langle -1, \xi_2, \ldots, \xi_{11}\rangle$ has finite index prime to 331 in E, so $\lambda(E) = U$. For an ideal $\mathfrak{a}$ of K, prime to the auxiliary primes, with $\mathfrak{a}^{331} = (\Gamma_{\mathfrak{a}})$, set $\varphi([\mathfrak{a}]) = \lambda(\Gamma_{\mathfrak{a}}) \bmod U$. This does not depend on the choice of $\Gamma_{\mathfrak{a}}$, it is additive, and it vanishes on principal ideals: if $\mathfrak{a} = (\delta)$, then $\Gamma_{\mathfrak{a}} = \varepsilon\delta^{331}$ with $\varepsilon \in E$, and δ is a unit at the auxiliary primes. Since every ideal class contains ideals prime to a given finite set of primes, φ is a homomorphism on the classes of order dividing 331.

Let $s = [P_{139}]_{331}$; it generates $\mathrm{Cl}_{331} \cong \mathbb{Z}/331$ [Gau26, Rem. 6.8(a)]. Write $[I]_{331} = c\,s$. The class $\mathrm{N}_{L/K}[I] = [\mathfrak{q}]^e$ has order dividing 331, and the norms of the components of $[I]$ of order 3 and 45 013 have orders prime to 331; by uniqueness of the primary decomposition in $\mathrm{Cl}(K)$, they vanish, and $\mathrm{N}_{L/K}[I] = c\,\mathrm{N}_{L/K}(s)$. Likewise $\mathrm{N}_{L/K}(s) = \mathrm{N}_{L/K}[P_{139}] = [\mathfrak{q}_{139}]^2$. Applying φ gives $e\,\lambda(\Gamma_{\mathfrak{q}}) \equiv 2c\,\lambda(\Gamma_{139}) \pmod{U}$. Since $\lambda(\Gamma_{139}) \notin U$ (rank 11), c is unique. □

Corollary 4.8. *For the 40 ideals listed in Table 3, in §§2–4 and in [Gau26], among them those of X_{139}, X_{599}, X_{4969}, X_{4049} and all the ideals of Proposition 4.5, the hours given by Proposition 4.7 agree with those of the census, in particular with Table 3. The identification of the 30 representatives of Table 3, and the twin and same-orbit relations of Proposition 4.5,*

therefore hold without GRH; the latter now have two proofs. For the layer, this settles question (Q5) of [Gau26]. [proved]

Justification. A PARI/GP script finds each $\Gamma_{\mathfrak{q}}$ with `bnfisprincipal` (under GRH), then checks without class group computations: the maximality of the order of K (`nfcertify`); the equality of ideals $(\Gamma_{\mathfrak{q}}) = \mathfrak{q}^{331}$, and that p is not an auxiliary prime; that the ξ_a are integral of norm ± 1; the ranks 10 and 11 modulo 331; and it solves the linear system of Proposition 4.7. It reports 97 checks and no failure, for 40 ideals: 36 ideals chosen above the primes concerned (for a prime of degree 2, the first factor of Φ_{115} found by PARI), and the four ideals P_{139}, P_{599} of [Gau26] and P_{4969}, P_{4049} of Proposition 2.1, exactly as given there; the 3-components are read from $4p = a^2 + 23b^2$. For these four ideals, $c = 1$ (by definition), 63, 41 and 53; for $P_{691}\sigma_{24}(P_{691})$, $c = 227$. A conjugate ideal changes c by an element of μ_{22}, not the hour: the two pairs of Proposition 4.5(a), (b) have the hours 8 and 13. The hours of all 40 ideals coincide with those computed under GRH. A second program, in Python, in exact integer and rational arithmetic and without PARI/GP, rechecks from the certificate file, against a fixed list of the 40 expected ideals: the primality of each p, the consistency of its type with e, the link of each $\mathfrak{q}$ with its ideal of L, the equalities $(\Gamma_{\mathfrak{q}}) = \mathfrak{q}^{331}$ (integrality, norm and valuations), the units, the ranks and the 40 values of c: 186 checks, no failure. Its first version, whose contract Astra found incomplete on October 17 (no primality test, no check of e against the type, no fixed list of targets, and floating-point square roots in bounds), gave the same 40 values. □

4.5. **The census of the layer.** For each of the 30 non-central orbits, a representative of small norm was sought among two families: prime ideals of degree 2 above $p \equiv 24 \pmod{115}$, and ideals $P\sigma_{24}(P)$ with P of degree 1 above $p \equiv 1 \pmod{115}$; in both cases $\mathrm{N}(I) = p^2$. The cost of a sweep grows like $\sqrt{\mathrm{N}(I)} = p$: the ratio of the values of $|\mathcal{U}|$ for X_{4969} and X_{139} is 35.96, against $4969/139 = 35.75$. A search up to $p = 300\,000$ (574 primes, 21 s) reaches all 30 orbits. Table 3 gives the cheapest representative of each orbit, by hour. The search used GRH to label the ideals; Corollary 4.8 certifies all the hours without GRH. The verdicts never depended on GRH.

A complementary search, for prime ideals of degree 1 whose class has zero 45 013-component (about one prime in 45 013; cost $\sqrt{p}$ only, since $\mathrm{N}(P) = p$), was carried out up to $p \approx 1.13 \cdot 10^9$ (648 833 primes $p \equiv 1 \pmod{115}$, four slices, about 69 hours of computation in total); it tests the vanishing of the 45 013-component in $\mathrm{Cl}(K_{44})$, under GRH. It found 9 primes of the layer, and improves only one orbit: hour 10 of the inner ring, by a prime above $p = 157\,146\,581$, of cost $\sqrt{p} \approx 12\,536$ instead of 17 389. For this prime, the label $c = 318$ and the hour are certified by Proposition 4.7 ($e = 1$), but the vanishing of the 45 013-component remains under GRH.

Filters of the night. On the night of October 16 to 17, the program `Rang88CertZ` was run as a filter on the representatives, in Table 3, of 21 of the 22 undecided orbits (all but that of P_{3359}, a partial survivor of a filter of [Gau26] with this same representative), on the author's computer: both sides in parallel, 5 threads each, stopped at the first hit and capped at 12 minutes. Every hit was checked exactly by a PARI/GP script that uses only the coordinates (u, y) of the hit: the basis generates the ideal (up to Galois conjugation for a prime of degree 2, which does not change the orbit), the Gram matrix recomputed from α is integral, even, unimodular and positive definite, and the vector has integer coordinates and norm 6. A rejection is counted only if this check succeeds entirely (exit code, final sentinel, complete count of checks, no error). Table 4 gives the outcome: 4 orbits are rejected, which is proved without GRH, and 17 are partial survivors, about which nothing is proved. Each surviving filter went through 4.3 to 10.9 million short projections, both sides together; in [Gau26, §6.3], the rejections of the same family occurred after 0.30 to 0.43 million short projections of side 1. [observed]

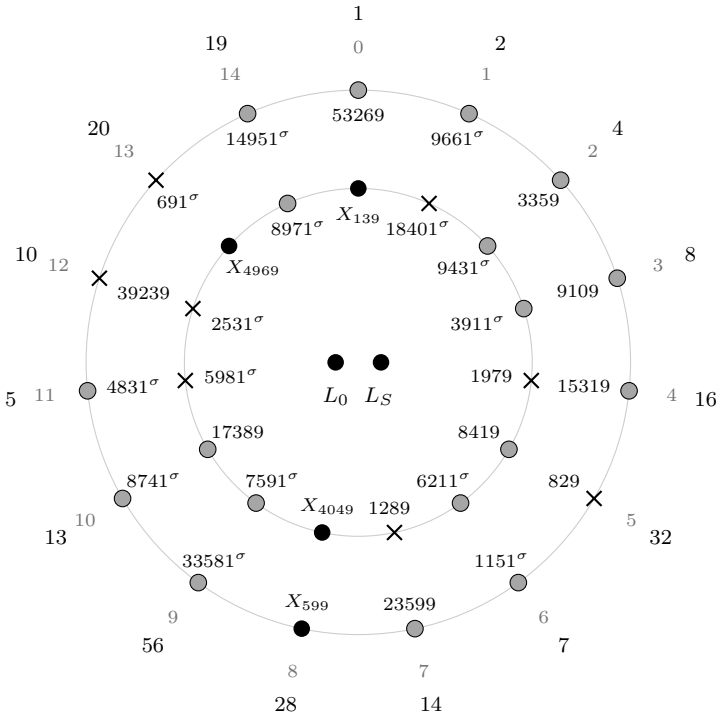


FIGURE 1. The σ_{24}-layer. Inner ring: the 15 orbits of support $\{3, 331\}$ (normalizer of order 460); outer ring: their 15 twins, of support $\{331\}$ (normalizer of order 920); centre: the twin pair (L_0, L_S). Outside: hour h (small) and smallest exponent k. Black disc: extremal (computer-assisted; [Sak26] for the centre); cross: a vector of norm 6 (proved); grey disc: partial survivor of a filter (of [Gau26] or of §4.5); open circle: not examined. A label p^σ denotes the orbit of $P\sigma_{24}(P)$, for a prime P of degree 1 above p; p alone, that of a prime of degree 2 above p. Hours: proved without GRH (Corollary 4.8).

4.6. **A heuristic: symmetry protects extremality.** The heuristic of [Gau26, (Q2)] (Siegel mean, Poisson law on the orbits of short vectors) gives a lattice whose normalizer of $\langle\zeta\rangle$ has order 230 a probability of about 4.4% of being extremal. If the vectors of norm ≤ 6 are distributed as in a lattice without extra symmetry, a normalizer of order $230m$ divides the number of their orbits by m, and the probability becomes about $0.044^{1/m}$: 21% for the inner ring ($m = 2$) and 46% for the outer ring ($m = 4$). [heuristic]

Before the filters of §4.5, the data were small but suggestive. In the inner ring, three of the five decided orbits are extremal; in the outer ring, one of three; among the generic orbits of size 88, none of the fifteen decided ideals of degree 1 is (the four rejections of [Gau26] and the eleven new rejections of the control group). With these probabilities, the 22 orbits of the layer then undecided would contain about $10 \times 0.21 + 12 \times 0.46 \approx 7.6$ further extremal lattices. [heuristic]

The filters of §4.5 give a first test. If the heuristic were right, and if a filter of this strength caught every non-extremal lattice, about 79% of the inner orbits and 54% of the outer orbits filtered would have been rejected. In fact 3 of the 10 inner orbits and 1 of the 11 outer orbits were rejected, whereas the filters of [Gau26] and of §3 had rejected 15 of the 17 generic ideals of degree 1 they examined. [observed]

Under these two hypotheses, and if the orbits behaved independently with these common probabilities, a deviation at least as large would have a probability of about 0.1% for the inner ring and 0.3% for the outer ring (binomial law). This number does not measure the

TABLE 3. The 30 non-central orbits of the σ_{24}-layer, by hour: cheapest known representative (p means $\mathrm{N}(I) = p^2$; (a) a prime of degree 2, (b) an ideal $P\sigma_{24}(P)$) and status. "ext.": extremal, computer-assisted; "rej.": vector of norm 6, proved; "part.": partial survivor of a filter. An asterisk marks the filters of the night of October 16 to 17 (Table 4).

		support $\{3, 331\}$ (inner ring)		support $\{331\}$ (outer ring)	
h	k	representative	status	representative	status
0	1	139 (a)	ext. (X_{139})	53 269 (a)	part.*
1	2	18 401 (b)	rej.*	9 661 (b)	part.*
2	4	9 431 (b)	part.*	3 359 (a)	part.
3	8	3 911 (b)	part.*	9 109 (a)	part.*
4	16	1 979 (a)	rej.	15 319 (a)	part.*
5	32	8 419 (a)	part.*	829 (a)	rej.
6	7	6 211 (b)	part.*	1 151 (b)	part.*
7	14	1 289 (a)	rej.	23 599 (a)	part.*
8	28	4 049 (a)	ext. (X_{4049})	599 (a)	ext. (X_{599})
9	56	7 591 (b)	part.*	33 581 (b)	part.*
10	13	17 389 (a)	part.*	8 741 (b)	part.*
11	5	5 981 (b)	rej.*	4 831 (b)	part. (5 659 (a)); part.*
12	10	2 531 (b)	rej.*	39 239 (a)	rej.*
13	20	4 969 (a)	ext. (X_{4969})	691 (b)	rej. (prop. 4.6)
14	19	8 971 (b)	part.*	14 951 (b)	part.*

probability that the hypotheses are false: it only says that such a deviation would be rare if they were right, and a fluctuation in a small sample is not excluded. Two explanations are possible, without excluding a fluctuation or dependences between orbits: extremal lattices may be more frequent in the layer than the heuristic predicts, or the short vectors of its non-extremal lattices may be harder for a filter to reach. Moreover, if the number of short projections grows like p, as the comparison of X_{139} and X_{4969} suggests, a filter of fixed duration, which goes through 4.3 to 10.9 million of them (Table 4), covers a fraction of the enumeration that decreases like $1/p$: the survival of the most expensive orbits says less. Only complete sweeps can decide. [heuristic]

With the representatives of Table 3, a cost of $0.0035\,p$ hours of a rented 32-core machine for a complete execution of both sides (calibrated on X_{4969}), a quick rejection for a non-extremal lattice and two executions for an extremal one, the expected cost of this census was about 830 hours before the filters of §4.5. For the 18 orbits still open, it is about 660 hours with the same probabilities, and about 1 700 hours if they are all extremal. Beyond the layer, the 1 023 orbits fixed by σ_{91} (support $\{45\,013\}$, normalizer of order 460) would contain about two hundred, and the 507 408 generic orbits, at 4.4%, more than twenty thousand. These numbers are not results: the model was compared with only two cases in rank 56, and it underestimated one of them. They say where to look. [heuristic]

5. OUTLOOK

(1) *The census of the layer.* Of the 22 orbits undecided before the filters of §4.5, 4 are now rejected. The other 18 can be settled by the same pipeline: each needs a complete sweep, stopped at the first hit for a non-extremal lattice, and two executions for an extremal one; the cheapest, that of $P_{1151}\sigma_{24}(P_{1151})$, would take about four hours of a rented 32-core machine per execution. By [Gau26, Rem. 2.6], each new extremal lattice of the layer has a normalizer of $\langle\zeta\rangle$ of order 460 or 920 in its automorphism group.

TABLE 4. Filters of the night of October 16 to 17, in the order of increasing cost. Projections: millions of short projections gone through, both sides together. For a rejection, the energies of the hit (short and fiber; sum 12), and the side on which it occurred.

h	ring	ideal	verdict	projections	time (s)	energies of the hit
6	outer	$P_{1151}\sigma_{24}(P_{1151})$	partial survivor	4.3	722	
12	inner	$P_{2531}\sigma_{24}(P_{2531})$	rejected (side 2)	0.2	71	$\frac{11913}{2531}+\frac{186878916}{5062^2}$
3	inner	$P_{3911}\sigma_{24}(P_{3911})$	partial survivor	6.0	723	
11	outer	$P_{4831}\sigma_{24}(P_{4831})$	partial survivor	8.3	722	
11	inner	$P_{5981}\sigma_{24}(P_{5981})$	rejected (side 1)	3.9	315	$\frac{28185}{5981}+\frac{1042775388}{11962^2}$
6	inner	$P_{6211}\sigma_{24}(P_{6211})$	partial survivor	7.1	720	
9	inner	$P_{7591}\sigma_{24}(P_{7591})$	partial survivor	6.5	722	
5	inner	P_{8419}	partial survivor	8.1	721	
10	outer	$P_{8741}\sigma_{24}(P_{8741})$	partial survivor	7.6	721	
14	inner	$P_{8971}\sigma_{24}(P_{8971})$	partial survivor	8.4	720	
3	outer	P_{9109}	partial survivor	9.3	721	
2	inner	$P_{9431}\sigma_{24}(P_{9431})$	partial survivor	7.8	724	
1	outer	$P_{9661}\sigma_{24}(P_{9661})$	partial survivor	7.5	720	
14	outer	$P_{14951}\sigma_{24}(P_{14951})$	partial survivor	10.1	723	
4	outer	P_{15319}	partial survivor	10.9	720	
10	inner	P_{17389}	partial survivor	8.7	721	
1	inner	$P_{18401}\sigma_{24}(P_{18401})$	rejected (side 1)	0.2	101	$\frac{85506}{18401}+\frac{9959062824}{36802^2}$
7	outer	P_{23599}	partial survivor	9.4	720	
9	outer	$P_{33581}\sigma_{24}(P_{33581})$	partial survivor	8.6	720	
12	outer	P_{39239}	rejected (side 2)	1.8	335	$\frac{164472}{39239}+\frac{48090690576}{78478^2}$
0	outer	P_{53269}	partial survivor	9.0	730	

(2) *Twins.* Theorem B shows that extremality is not a twin invariant. Four twin pairs are now decided: two pairs of extremal lattices, (L_0, L_S) and (X_{4049}, X_{599}); one pair of non-extremal lattices, at hour 12 ($P_{2531}\sigma_{24}(P_{2531})$ and P_{39239}, rejected by the filters of §4.5); and the discordant pair of Theorem B. Whether extremality is *correlated* between twins is open. The filters leave five pairs with one member rejected and the other a partial survivor: the inner member is rejected at hours 1, 4, 7 and 11, the outer one at hour 5. The twin of X_{139}, P_{53269}, survived a filter that covered only a small fraction of its enumeration. Complete sweeps of these survivors would show whether discordant pairs are frequent.

(3) *The σ_{91}-layer.* Proposition 4.1 has an analogue for the classes fixed by σ_{91}: they form the cyclic group $\mathrm{Cl}_{45\,013}$, with 1 023 non-trivial orbits of size 44. Its census requires cheap representatives, which a search of the kind of §4.5 would provide.

(4) *Consolidation.* An independent implementation of the enumeration remains the main missing step (question (Q1) of [Gau26]); the recount of §3 by `qfminim` is a first, non-certifying, step in that direction.

DATA, METHODS AND DECLARATIONS

Data. The input files, receipts, logs, verifiers and certificates of this addendum will be added to the public repository of [Gau26], in the folders `addendum` (twin certificates, labels without GRH, filters of §4.5), `verification_P4969`, `verification_P4049`, `calcul_externe_resultats_20261015` and `contre_audit_20261016`. The computations

used PARI/GP 2.15.4, Python 3 in exact integer and rational arithmetic, and Java (OpenJDK 11, 17 and 21). The sweeps ran on rented Scaleway instances POP2-HC-32C-64G.

Use of artificial intelligence. As for [Gau26], the author was helped by two assistants, Claude Opus 5.5 (Anthropic) and Astra (OpenAI), who are not authors. The text, the programs and the certificates were written, and the computations conducted, with Claude Opus 5.5. Astra designed part of the audit plan, had the records listed in §3.7 attacked and reread, and found the defects reported there. On October 17, Astra also read, without executing them, the argument of Proposition 4.7 with its two verifiers, and the launcher, the witness verifier and the audit of the filters of §4.5. It confirmed the argument. It required the hypothesis that $\mathfrak{q}$ be prime to the auxiliary primes, and it found gaps in the contract of the first Python verifier (no primality test, no check of e against the type, no fixed list of targets, floating-point square roots), a defect in the first launcher, which recorded a hit as a rejection whatever the outcome of the exact check, and a contract of the witness verifier that was too broad, now restricted to the two forms of ideals actually used. All were corrected, and every rejection was re-verified. Astra also read the twin and same-orbit certificates and the witness of norm 6 of $P_{691}\sigma_{24}(P_{691})$, without executing them: it found no mathematical defect in the positive chain of Theorem B, and asked for the four elements δ of the same-orbit certificates to be exported, which was done. The census and the other results of §4 have not been reviewed by Astra.

The author takes scientific responsibility for the entire content. This addendum has not been reviewed by a human specialist.

References

[Gau26] I. Gauvin, *Even unimodular ideal lattices of rank* 88 *over* $\mathbb{Q}(\zeta_{115})$, preprint, October 15, 2026 (arXiv identifier pending); French original: *Réseaux idéaux pairs unimodulaires de rang* 88 *sur* $\mathbb{Q}(\zeta_{115})$. Repository: https://github.com/Katabolanga/rang88-reseaux-ideaux.

[NS] G. Nebe, N. J. A. Sloane, *A Catalogue of Lattices*, https://www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/.

[Sak26] R. Sakamoto, *Two extremal even unimodular lattices of rank* 88, preprint arXiv 2609.15633, version 2 of September 27, 2026.